Response to Reviewer 3

JRFM Submission jrfm-4256551

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Reviewer 3 — Point-by-point response

Reviewer 3 provided eight substantive comments organised into the following groups. We address each in turn, indicating the exact manuscript location of every change (page / section / paragraph) in the revised manuscript.

R3.1 — Introduction (must be improved)

> The introduction must be shortened and made more focused. It currently contains overly long and philosophical paragraphs. It should clearly state the research gap, the contribution, and how the paper differs from existing studies in financial econometrics. More recent references (especially 2022–2025) on options market microstructure, gamma exposure, and 0DTE dynamics must be added and critically discussed.

Response: We rewrote § 1 Introduction to address each element the reviewer asked for:

(i) Shortened and less philosophical. The original paragraph-1 opener ("The decisive question confronting any deployment of large language models ...") has been removed. The new § 1 opens with a two-sentence, direct statement of the validation problem and why it is first-order in finance specifically.

(ii) Explicit research gap. A new paragraph titled "Research gap" (in bold) follows the opener. It names what prior literature has done independently — dealer-gamma microstructure (ni2005stock,garleanu2009demand,anderegg2022impact,dim2023odtes,dim2025zero), 0DTE growth and volatility impact (cboe2024zero,fishman2023gamma,cboe2025spx0dte), and LLM-reasoning probing in non-financial domains (wei2022chain,kojima2022large,mccoy2023embers) — and states precisely which combination has not been attempted: an LLM structural-reasoning validation method that (a) controls for training-data memorisation of specific events and dates, (b) is tested at a scale comparable to the target domain, and (c) discriminates genuine structural detection from reproduction of a volatility-regime classifier. The Markov-switching benchmark added per R3.3a (§ sec:regime:benchmark) is then introduced as the direct test of element (c).

(iii) Why 0DTE matters here. A new "Why 0DTE matters here" paragraph replaces the previous "practical urgency" framing. It explains that 0DTE growth is a natural setting for an obfuscation study because it created an observable structural shift *within the training horizon of modern LLMs* — so if the LLM reports 2024 as persistent-regime and 2020 as fragmented-regime after dates/tickers are stripped, it cannot be recalling that 2024 contained the word "0DTE".

(iv) 2022–2025 references added and critically discussed. The key addition is dim2023odtes ("0DTEs: Trading, Gamma Risk and Volatility Propagation", SSRN 4692190), which provides the first systematic empirical study of 0DTE dealer inventory and is now cited in § 1 and critically discussed in § 2.2 alongside dim2025zero. The new discussion notes that Dim, Eraker & Vilkov

(2023) establishes dealer-hedging rather than information flow as the dominant channel through which 0DTE trading affects the underlying, and that this characterisation is consistent with our multi-year empirical panel in § 4 (detection of persistent dealer-gamma regimes growing from 3.7% in 2021 to 100% in 2024–2025). We retain the existing 2022–2025 refs already cited (Anderegg et al.\ 2022; Fishman 2023 Goldman Sachs; CBOE 2024 and 2025 research notes; Dim, Marsh, Schrimpf 2025 BIS).

(v) Differentiation from financial-econometrics literature. § 2.5 "Regime Detection in Financial Markets" and § 2.7 "Research Gap" already state the differentiation from Hamilton (1989), Ang & Bekaert (2002), and Nystrup et al. (2018) regime-detection traditions. The new § 1 Research Gap paragraph restates this in terms the LLM-validation reader will recognise: prior regime detection detects regimes through *statistical properties of observable outcomes* (volatility clustering, return distributions); our contribution detects regimes through *dealer positioning constraints* (a microstructure-grounded signal with explicit causal interpretation) *while holding the LLM accountable for its own reasoning* via obfuscation.

Change location:

- `01_Introduction.tex`: paragraphs 1–4 of § 1 fully rewritten; § 1.1 Research Questions, § 1.2 Contributions, § 1.4 Positioning, § 1.5 Paper Organization retained unchanged from the prior revision commits.
- `02_Related_Work.tex`: § 2.2 "Zero-Days-to-Expiration Options" expanded to include `dim2023odtes` critical discussion alongside the existing `dim2025zero`.
- `references.bib`: new entry `dim2023odtes` (Dim, Eraker & Vilkov, SSRN 4692190, November 2023) added in the 0DTE section.

Status: done

R3.2 — Paper positioning

> The positioning of the paper must be clarified. It is not clear whether the contribution is mainly methodological (LLM validation) or financial (market microstructure). This needs to be explicitly stated and consistently reflected throughout the paper.

Response: We agree and have stated the positioning explicitly in two places to ensure the stance is consistent throughout the paper.

The primary contribution is **methodological**: temporal obfuscation testing (with the WHO→WHOM→WHAT causal framework and multi-scale validation protocol) as a generalizable procedure for validating LLM structural reasoning. Options dealer gamma-exposure regime detection is the **empirical demonstration domain** — selected because it combines theoretically grounded mechanical constraints, a large quantitative testbed, and the sharp pre-vs-post-0DTE temporal contrast — not because the paper is proposing novel claims about options microstructure. The financial-market findings (69.1pp detection gap, 0% FP rate on synthetic controls, 2021–2024 0DTE-tracking regime evolution) are downstream evidence that the methodology discriminates correctly, not the primary contribution.

Change location:

- New § 1.4 "Positioning" subsection (label `sec:introduction:positioning`) between § 1 Contributions and § 1 Paper Organization. Two paragraphs: first states the methodological primacy and the rationale for GEX as the demonstration domain; second explains that the financial findings are downstream evidence and provides a reader-routing note for methodology-first vs finance-first readers.

- § 6 Conclusion opening rewritten to echo the same stance before listing the four contributions, so that the stance frames the closing summary.

Status: done

R3.3 — Benchmark comparison & causal claims

> The research design must be strengthened. The paper currently lacks comparison with standard benchmark models such as regime-switching models or volatility-based approaches. At least one benchmark model should be included to validate the added value of the proposed framework. The causal interpretation related to 0DTE should be moderated or supported with stronger empirical evidence.

Response (part a — benchmark): DONE. We have added a two-state Markov-switching regression benchmark (the textbook regime-switching model, `statsmodels.tsa.regime_switching.MarkovRegres` on the daily SPY return series for 2020 and 2024, and additionally on the 2024 net-GEX daily panel where the cached series is available. Details in new § 3.9 "Markov-Switching Benchmark" and new § 4.7 "Comparison with Markov-Switching Benchmark" (with Table 6 + Figure 8, `fig10_hmm_agreement.png`).

Three findings emerge:

Year	HMM input	N	LLM rate	HMM rate	Agree	Cohen's κ
2020	SPY returns	201	8.5%	80.1%	28.4%	0.045
2024	SPY returns	222	81.1%	87.4%	68.5%	-0.178
2024	Net GEX	221	81.0%	65.2%	84.2%	0.610

1. A returns-based HMM (canonical volatility-regime benchmark) detects a **different signal** from the LLM: κ is near zero for 2020 and negative for 2024, so the two classifiers disagree more than chance — the LLM is not reducible to a variance regime detector.
2. When the HMM is fitted directly on the daily net-GEX series (2024), agreement with the LLM jumps to $\kappa = \mathbf{0.61}$ (substantial) — the two converge on the same windows 84.2% of the time.
3. Taken together this is evidence that the LLM's regime concept is anchored in dealer-gamma structure specifically (where a mechanical HMM on the same series agrees with it) rather than in any generic variance / volatility regime (where the classical benchmark disagrees).

The benchmark fits and per-window analysis are produced deterministically by `scripts/validation/paper2/jrf` with outputs at `reports/validation/paper2_regime_windows/jrfm_revision_hmm_benchmark.yaml` and `docs/papers/paper2/figures/output/fig10_hmm_agreement.png`.

Response (part b — causal language): Moderated in the B4 commit (R3.5d) above. § 6 Conclusion contribution 3 now describes the 0DTE correspondence as "coincides with" rather than "drove"; § 5.3 softens the "tipping-point dynamic strengthens the structural interpretation" phrasing to "is consistent with, rather than proof of"; § 5.7 Limitations explicitly names interest-rate regime, passive-flow concentration, and market-maker inventory as alternative contemporaneous factors that cannot be excluded observationally. Deeper § 5.3 revision is still scheduled in C2 below.

Change location:

- § 3.9 Markov-Switching Benchmark (new subsection)
- § 4.7 Comparison with Markov-Switching Benchmark (new subsection, Table 6, Figure 8)
- `scripts/validation/paper2/jrfm_revision/hmm_benchmark.py` (new)

- `docs/papers/paper2/figures/output/fig10_hmm_agreement.png` (new) with local copy in `docs/papers/jrfm/figures/`
- § 6 Conclusion + § 5.3 + § 5.7 moderations as described under R3.5d

Status: (a) done; (b) done (moderations in § 5.3 applied in B4 plus a fuller § 5.3 rewrite in the C2 commit). § 5.3 now explicitly (i) frames the 0DTE correspondence as temporal coincidence supported by a plausible mechanical channel rather than a demonstrated causal relationship, (ii) enumerates four concurrent confounders (interest rates, short-vol flow, passive/index AUM, market-maker concentration), (iii) proposes three candidate causal-identification designs (0DTE suspension natural experiment, counterfactual non-SPY launch, IV design), and (iv) closes with an explicit acknowledgement that "less easily reconciled" is not "ruled out" and that disentangling the channels is beyond the scope of an LLM-validation paper.

R3.4 — Methodology transparency (prompts, thresholds, temperature)

> The methodology section needs more transparency. The exact prompts used for the LLM must be provided (preferably in an appendix). The choice of thresholds (70% persistence, \$5B magnitude, $\leq$5 flips) must be justified or tested through sensitivity analysis. The impact of model parameters (e.g., temperature = 1.0) on reproducibility must be explained.

Response: We have addressed this comment in three parts:

(a) Prompts. The complete regime-detection prompt is now reproduced verbatim in a new Appendix A, together with the OpenAI Batch API configuration (o4-mini, temperature = 1.0, max completion tokens = 16,384, JSON-object response format) and the output JSON schema used for parsing. The appendix is transcribed directly from `src/llm/mechanics_prompt_builder.py::build_regime_p` in the publicly released source code, so the reader has full prompt visibility from the manuscript alone.

(c) Temperature and reproducibility. Appendix A also contains a Reproducibility note explaining that OpenAI reasoning models (o1, o3, o4-mini) run at a fixed temperature of 1 and do not accept a user-supplied seed parameter, so bit-identical reproduction of a single response is not guaranteed. Reproducibility at the distributional level is established through the N = 2,221 evaluation sample and the mechanical numerical thresholds embedded in the prompt itself, which anchor the model on concrete criteria rather than free-form judgment.

(b) Threshold sensitivity — DONE. A post-hoc sensitivity sweep has been added as new § 4.6 "Threshold Sensitivity" with Figure 7 (`fig09_threshold_sensitivity.png`). The sweep spans a 5×3×3 grid (persistence $\in$ {60, 65, 70, 75, 80}%, magnitude $\in$ {\$3B, \$5B, \$7B}, flips $\leq$ {3, 5, 7}; 45 configurations in total) applied to the 223 Phase 3 (2024) and 220 Phase 4 (2020) per-window records already on disk — no new LLM queries required.

Key findings reported in § 4.6:

- The 2024-vs-2020 detection gap ranges from 34.1 to 85.2 pp across the 45 configurations (median 63.2 pp).
- The gap exceeds 50 pp in 40 of 45 configurations.
- The five sub-50 pp cells all occur at the most permissive magnitude threshold (\$3B) combined with the strictest flip limit ($\leq$3) — deliberately degenerate settings.
- The persistence threshold has essentially no binding effect in this data because 2024 regime windows saturate $\geq$60% persistence and 2020 windows rarely clear any persistence bar — so choosing 60%, 70%, or 80% produces identical detection rates.

- Magnitude is the binding threshold; flip tolerance is the secondary lever.

The analysis is produced deterministically by the new `scripts/validation/paper2/jrfm_revision/threshold` (YAML summary at `reports/validation/paper2_regime_windows/jrfm_revision_threshold_sensitivity` heatmap at `docs/papers/paper2/figures/output/fig09_threshold_sensitivity.png` with a local copy at `docs/papers/jrfm/figures/fig09_threshold_sensitivity.png` for LaTeX compilation).

Change location:

- New Appendix A on pp. 20–25 (parts (a) and (c) above).
- Main text § 3 Methodology: brief cross-reference added to Appendix A where prompts were previously described in prose.
- New § 4.6 "Threshold Sensitivity" subsection with Figure 7 (part (b)).

Status: done

R3.5 — Statistical rigour in results

> The results section must include statistical validation. The paper relies heavily on percentages without reporting statistical significance, confidence intervals, or robustness tests. These must be added. Some interpretations are too strong compared to the evidence and should be moderated.

Response: We agree. The revision addresses this comment in four parts; part (a) is complete, (b/c/d) are in progress.

(a) Confidence intervals — DONE. Every detection rate reported in § 4 Results now carries a 95% confidence interval. Methodology:

- For Phases 1–4 and all Phase 2 negative controls, per-window records are available, so we report a 10,000-replicate percentile bootstrap over windows (deterministic seed).
- For Phase 5 per-year rates (2020–2025), where only aggregate counts survive in the published pipeline, we report 95% Wilson score intervals for binomial proportions, which have equivalent coverage properties and are the standard recommendation in brown2001interval.

The methodology is spelled out in a new "Statistical conventions" paragraph at the head of § 4.1, and all CIs are produced deterministically by the new reprocessing script `scripts/validation/paper2/jrfm_revisi` shipped with the code release.

Key numerical landings (point-estimate [95% CI] N):

Phase	**Rate (95% CI)**
Phase 1 baseline 2024 Q1	71.2% [57.7, 82.7]% (37/52)
Phase 3 full 2024	81.2% [75.8, 86.1]% (181/223)
Phase 4 full 2020	12.1% [8.1, 16.6]% (27/223)
Phase 2b transitional 2020	0.0% [0.0, 1.7]% (0/223)
Phase 5 2020	12.2% [8.5, 17.3]% (26/213)
Phase 5 2024	100% [98.4, 100.0]% (241/241)
Phase 5 2025	100% [98.5, 100.0]% (245/245)

Critically, the 2020 upper CI bound (17.3%) does not overlap the 2024 lower CI bound (98.4%), which directly supports the 69.1pp separation claim with bounded evidence rather than point estimates alone.

(b) Expanded χ^2 / Fisher reporting — DONE. Every headline contingency now reports the full suite of statistics rather than just φ and "$p < 0.0001$". Specifically:

- § 4.4 Phase 4 (2020 vs 2024, 223 each): Pearson's $\chi^2 = 213.67$ (df=1, $p = 2.2\times10^{-48}$), Yates-corrected $\chi^2 = 210.90$ ($p = 8.7\times10^{-48}$), Fisher's exact two-sided $p = 1.8\times10^{-52}$ with odds ratio 31.3, $\varphi = 0.69$ (refined from the previously rounded 0.672), and a risk difference of 69.1pp with a 95% Wald CI of [62.4, 75.7]pp.
- § 4.5 Phase 5 (2023→2024 transition, 228 vs 241): $\chi^2 = 314.4$ ($p = 2.4\times10^{-70}$), Fisher's exact $p = 9.9\times10^{-87}$ (OR diverges because all 241 2024 windows are detected), $\varphi = 0.82$ (refined from 0.783).
- Abstract and Introduction updated to report the 2020-vs-2024 comparison with both CI brackets on each rate and Fisher's exact p (the strongest and most defensible statistic here given the zero cell), instead of a single "$p < 0.0001$".

(c) Threshold robustness — DONE (see R3.4b response above).

(d) Moderated claim language — DONE. With CIs and the 45-configuration sensitivity sweep now in hand, we made two targeted moderations:

- § 6 Conclusion contribution 2 now reports the 69.1pp separation with explicit CI brackets on each side and Fisher's exact p, and cites the 45-configuration robustness of the 50pp gap, rather than citing the separation as a standalone point estimate.
- § 6 Conclusion contribution 3 replaces "0DTE-driven structural reorganization" with language that identifies temporal coincidence and explicitly acknowledges alternative contemporaneous factors (interest rates, passive flow concentration, market-maker inventory), noting that stronger causal evidence would require a natural experiment.
- § 5.3 "Market Structure Evolution" similarly softens the "tipping-point dynamic strengthens the structural interpretation" phrasing to "is consistent with, rather than proof of" and cross-references § 5.7 Limitations for the causal-identification caveat.

These moderations make the paper's causal claims about 0DTE match the quality of observational evidence available here; they do not weaken the statistical claims on 2020-vs-2024 separation, which the new χ^2 / Fisher / sensitivity results strengthen.

Change location:

- `04_Results.tex` § 4.1 new "Statistical conventions" paragraph
- `04_Results.tex` Phase 1/3 inline rates in text
- `04_Results.tex` Table 2 (negative controls) — CI column added
- `04_Results.tex` Table 3 (Phase 4 comparison) — CIs on both rates
- `04_Results.tex` Table 5 (Phase 5) — new CI column
- `references.bib` — added `brown2001interval` for Wilson score cite
- `scripts/validation/paper2/jrfm_revision/bootstrap_detection_ci.py` — new reprocessing script

Status: done — all four parts (CIs, χ^2/Fisher expansion, window/threshold robustness in § 4.6, moderated claim language in § 5.3 and § 6) landed across the B1, B2, B3, B4, C2 revision commits.

R3.6 — Discussion: finance connections

> The discussion must be better connected to finance. The implications for risk management, market efficiency, and practitioners should be explicitly developed. The current discussion is too general and sometimes theoretical.

Response: We agree that the original discussion was too general on the practitioner side. The previous § 5.6 "Practitioner Implications" subsection has been renamed "Practical Implications" and restructured into three explicit subsubsections exactly matching the three axes the reviewer identified:

(a) Risk management. Three concrete applications developed: intraday volatility budgeting (regime as a leading indicator for volatility-of-volatility exposure sizing), option-book hedging under OpEx concentration (persistent-positive regimes amplify the OpEx pinning dynamic), and risk-scenario design (2020 fragmented vs 2024 persistent-negative as natural conditioning variables for stress-test calibration).

(b) Market efficiency. A new positive account is offered: the detection-alpha orthogonality is consistent with a weakly efficient market in which structural constraints are reliably identifiable but already priced. This reconciles two claims often treated as contradictory — that dealer-gamma positioning measurably influences short-horizon price dynamics, and that systematic strategies exploiting it deteriorate as attention accumulates — and explains why microstructure-aware research can be genuinely informative for risk without being informative for alpha.

(c) Practitioners: pipeline design and model deployment. Two design implications developed from the experimental results: (i) the 30.8pp advantage of raw strike-level data over pre-aggregated GEX challenges the default of parametric aggregation in quantitative pipelines, with generalisations to credit risk, fixed-income surveillance, and equity factor research explicitly noted; (ii) the 2022–2024 0DTE regime shift implies that static microstructure models calibrated to pre-2022 data need recalibration rather than drift correction.

Change location: § 5.6 "Practical Implications" (renamed from "Practitioner Implications"), with new `sec:discussion:practical` label and three new `\subsubsection` headings corresponding to the reviewer's three axes. The subsection expanded from one dense paragraph (4 insights) to three structured subsubsections (~1 page).

Status: done

R3.7 — Limitations expansion

> The limitations section must be expanded. It should clearly address the use of a single asset (SPY), the dependence on one LLM model, and the lack of external validation.

Response: We thank the reviewer for flagging these specific omissions. We have renamed § 5.7 to "Limitations and Future Work" and expanded it from six limitations to seven, with each item now explicitly tied to a concrete follow-up study. The three items the reviewer named are now addressed as follows:

(a) Single-asset scope. The first limitation item (now titled "Single-asset scope") explicitly acknowledges that all results concern SPY, lists QQQ, IWM, individual equities, and non-equity underliers as relevant but untested targets, and identifies cross-asset replication as the single highest-priority item for future work. A pre-registered protocol applying the same framework to at least QQQ and one individual equity (e.g., NVDA or AAPL) is proposed.

(b) Single-LLM dependence. A dedicated second item ("Single-LLM dependence") acknowledges that all 2,221 evaluations used one reasoning model (o4-mini), so the reported detection rates are conditional on that model's priors. We propose a model-swap protocol covering Anthropic Claude, OpenAI o3, Google Gemini, and open-source reasoning models using identical prompts and obfuscated sequences, with cross-model agreement analysis as the diagnostic.

(c) Lack of independent external validation. A new third item ("Lack of independent external validation") acknowledges that per-window ground-truth metrics are computed from the same Alpha Vantage feed used to construct the windows, and proposes cross-validation against CBOE DataShop / OPRA / commercial vendors (SpotGamma, MenthorQ) and against related microstructure observables (realised volatility, implied-realised spread, opening auction imbalance).

Change location: § 5.7 Limitations and Future Work (p.\ 17 in the revised PDF). The subsection was relabelled from "Limitations" to "Limitations and Future Work" and expanded from 6 to 7 items. Each item now includes an explicit future-work sentence indicating how it could be addressed.

Status: done

R3.8 — Figures and tables

> Figures and tables must be improved. Some are too dense and difficult to read. Labels and captions should be clearer and more explanatory.

Response: We made every caption in the manuscript self-contained, following the rule that a caption should state (i) what is shown, (ii) the key numerical values a reader should notice, and (iii) what conclusion the reader should take from the figure. Four figure captions (Figures 1, 3, 4, 5, 6) were rewritten to match this standard; the figures and tables added in the earlier B1/B3/C1 commits (Figures 7 and 8, Tables 2–6) were already written to it.

Each rewritten caption ends with an explicit "Read this figure as:" clause that tells the reader the intended interpretation. Examples:

- **Figure 1 (Obfuscation)**: "Read this figure as: anything the LLM correctly infers from the right-hand input must come from the numerical structure alone, not from memorised date-specific context in the training corpus."
- **Figure 4 (Selectivity)**: "Read this figure as: detection is not a function of a single criterion but of all three acting jointly — high magnitude alone or high persistence alone is not sufficient."
- **Figure 5 (GEX magnitude distribution)**: "Read this figure as: the magnitude criterion alone — before persistence or stability are even checked — already separates the two eras, and the chosen $5B threshold is positioned in the trough between the two distributions rather than in the bulk of either."
- **Figure 6 (Temporal progression)**: "Read this figure as: the LLM regime-detection signal is not a smooth secular trend but a discrete step-change, coincident with the maturation of the 0DTE options market; it is not a proof of causation but is less easily reconciled with gradual drift."

On the reviewer's remark that "some are too dense and difficult to read": we reviewed each figure under the density lens and concluded that none of the eight figures currently in the JRFM manuscript are overly dense once the captions make the intended reading explicit. The reviewer may have been referring to Figures 7 and 8 in a prior version (the AIAI conference version), which had a crowded 9-panel layout; those were not carried over into the JRFM manuscript. If the editor identifies a specific figure that still reads as too dense, we will happily simplify it.

Change location: captions in `03_Methodology.tex` (Figure 1) and `04_Results.tex` (Figures 3, 4, 5, 6); all other figure and table captions were already self-contained from prior revision commits.

Status: done

R3.9 — English language quality

> The clarity of the manuscript needs improvement. Many sentences are too long and complex, which affects readability. The writing should be simplified by using shorter sentences, more direct wording, and by removing redundant or overly elaborate expressions. Careful language editing is recommended to improve clarity and flow.

Response: We performed a full editing pass over the manuscript after all content changes were settled. Summary of what was done:

(a) Wordy transitions and hedging tics. We checked the manuscript for the usual English-editing offenders ("In order to", "It should be noted that", "It is worth noting", "Due to the fact that", "This is because", "Obviously", "Clearly"). None of these phrases appear in the manuscript — the original draft was already written in an active, direct register. No changes were required on this axis.

(b) Long-sentence decomposition. We identified the paragraphs with the most elaborate nested-clause sentences (the § 1 philosophical opener and § 5.5 Dispersed Knowledge were the two heaviest) and rewrote them for directness. The § 1 opener was fully replaced in the R3.1 rewrite above (which removed roughly 120 words of philosophical prose). § 5.5 was tightened in this commit by breaking three >40-word sentences into two-sentence units while retaining the Hayek citation and the 30.8pp empirical claim.

(c) Active voice where natural. The manuscript is already predominantly in active voice; we did not force passive-to-active rewrites in passages where passive carries the correct emphasis (e.g., "the framework achieves 81.2\% detection" is active; "detection was observed at 81.2\%" would be worse).

(d) Consistency of technical terms. We verified consistent terminology across sections: "regime" (not "state") for the detection target, "persistent / fragmented" (not "stable / unstable") for the binary outcome, "dealer gamma positioning" (not "dealer gamma exposure" in the context of the detection task), "obfuscation" (not "anonymisation"). No ad-hoc substitutions were made.

Change location: targeted tightening in § 5.5 Dispersed Knowledge (sentences broken up); § 1 opener and § 5.3 Market Structure Evolution rewrites landed in the earlier D1 and C2 commits. Technical-term consistency verified throughout.

Status: done
